

EVANSVILLE, IN, USA

WMO: 724320

Lat: 38.044N

Lon: 87.521W

Elev: 400

StdP: 14.48

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93817

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.3	13.8	-1.6	5.1	11.1	3.4	6.7	15.6	24.5	39.6	21.4	39.7	6.3	320	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	93.7	75.8	91.4	75.4	89.4	74.6	79.1	89.6	77.9	87.9	76.8	86.0	8.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.1	138.4	84.9	75.0	133.0	83.6	73.9	128.1	82.3	42.7	89.7	41.5	88.0	40.3	86.0	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.4	18.1	16.2	1.5	97.1	6.3	3.0	-3.0	99.3	-6.7	101.1	-10.3	102.8	-14.9	105.0		
			0.6	81.3	6.2	1.2	-3.8	82.2	-7.4	82.9	-10.8	83.6	-15.3	84.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	33.4	37.4	46.4	57.3	66.8	75.3	78.1	77.1	70.3	58.7	46.3	37.5
	DBStd	18.07	11.77	11.09	10.54	9.10	7.75	5.69	4.73	4.81	6.96	8.80	9.52	10.42
	HDD50	1742	530	375	196	35	1	0	0	0	0	21	180	404
	HDD65	4353	980	774	580	260	75	5	0	1	30	236	561	851
	CDD50	4352	15	21	85	253	522	758	870	841	609	290	70	18
	CDD65	1489	0	0	5	28	131	313	405	376	189	40	2	0
	CDH74	14401	0	1	37	241	1126	3052	4090	3676	1787	379	12	0
	CDH80	5612	0	0	2	28	314	1216	1733	1543	686	89	1	0
(14) Wind	WSAvg	6.7	8.1	8.1	8.1	8.1	6.8	6.1	5.3	4.9	5.0	6.0	6.9	7.2
(15) Precipitation	PrecAvg	44.6	2.9	3.2	4.6	4.6	4.9	3.8	4.1	3.0	3.2	3.0	3.9	3.8
	PrecMax	69.9	9.1	10.1	12.8	11.8	13.5	7.6	9.2	8.5	8.7	8.2	8.5	8.2
	PrecMin	27.9	0.5	0.6	1.3	1.1	0.9	0.2	0.2	0.2	0.2	0.0	0.5	0.6
	PrecStd	8.3	1.7	2.0	2.4	2.6	2.8	1.9	2.1	2.0	2.2	1.8	2.0	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.0	70.4	78.5	83.2	89.5	94.8	97.7	97.1	94.2	86.5	75.5	68.0
		MCWB	60.2	60.3	63.3	66.3	72.5	76.2	78.3	76.6	71.7	70.2	63.4	61.5
	2%	DB	61.4	65.1	73.4	80.2	86.2	91.7	93.2	93.5	90.7	82.4	70.3	62.4
		MCWB	57.6	57.1	60.9	65.2	70.6	75.0	77.3	75.9	72.2	68.0	60.5	57.5
	5%	DB	57.0	60.2	68.7	76.7	83.7	89.5	90.9	90.5	87.5	78.5	66.2	57.8
		MCWB	53.5	53.5	58.2	63.3	69.6	74.2	76.7	75.2	71.4	65.1	58.3	53.2
	10%	DB	51.6	55.3	64.0	73.2	80.8	87.1	88.6	88.0	83.8	74.1	62.1	53.5
		MCWB	47.0	49.0	54.9	61.4	68.5	73.2	75.5	74.1	69.4	63.4	55.2	49.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.4	62.5	65.4	70.3	75.4	79.1	81.0	80.5	76.5	73.3	66.2	62.7
		MCDB	64.4	68.3	75.9	79.6	84.3	90.4	91.0	91.5	87.0	82.8	71.6	67.2
	2%	WB	58.3	58.9	62.5	67.7	73.4	77.7	79.4	78.6	75.0	70.7	63.0	58.4
		MCDB	61.3	63.4	70.6	76.6	82.6	88.3	89.9	89.0	85.2	78.6	68.5	61.6
	5%	WB	54.0	54.8	59.7	65.5	71.7	76.4	78.3	77.3	73.5	67.9	59.6	53.9
		MCDB	56.8	59.4	66.9	73.4	80.5	86.3	88.2	87.1	82.9	74.9	65.0	57.0
	10%	WB	47.3	49.2	56.2	63.3	70.0	75.0	77.2	76.1	72.0	65.1	56.0	49.4
		MCDB	51.0	54.8	63.1	70.9	77.9	84.2	86.4	84.9	80.1	72.4	61.1	53.0
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	17.0	18.9	20.9	19.5	19.0	18.5	19.6	22.0	21.4	18.5	15.0
		MCDBR	19.5	24.0	24.4	24.0	22.0	21.2	20.8	22.2	24.9	25.3	23.4	19.3
	5% WB	MCWBR	16.9	17.9	14.8	13.2	10.0	8.3	8.1	8.3	9.4	12.2	15.8	15.7
		MCWBR	18.2	21.6	21.6	20.2	18.8	18.6	18.3	18.9	20.2	20.9	20.1	17.9
(40) Clear-Sky Solar Irradiance	taub	taub	0.315	0.327	0.367	0.408	0.440	0.451	0.474	0.481	0.417	0.362	0.329	0.310
		taud	2.455	2.399	2.352	2.242	2.203	2.195	2.141	2.111	2.281	2.429	2.469	2.470
	Ebn at Noon	Ebn at Noon	275	286	283	277	268	265	257	252	263	269	267	268
		Edh at Noon	26	31	36	43	46	46	48	49	39	30	26	24
(44) All-Sky Solar Radiation	RadAvg	RadAvg	625	851	1188	1571	1763	2018	1938	1783	1479	1092	744	538
	RadStd	RadStd	68	99	109	89	153	127	123	95	106	115	80	50

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+194	+140

Nomenclature: See separate page